

3.2 Ecosystems Under Threat

1. How People Use and Modify Ecosystems

Every ecosystem on Earth is used by people to obtain **food, energy and water**. Use is not automatically damaging — traditional pastoralism and shifting cultivation have operated inside savanna and rainforest ecosystems for centuries. Damage occurs when the rate of extraction exceeds the rate at which the ecosystem can recover, or when the structure of the ecosystem is changed so far that its cycles and flows stop working.

Resource	How ecosystems are used	How this modifies the ecosystem
Food	Clearing land for crops and pasture; grazing livestock; hunting and fishing; gathering wild foods	Replaces a diverse plant community with a single species (monoculture); removes herbivores or predators from the food web; strips nutrients from the soil with each harvest
Energy	Firewood and charcoal collection; biofuel plantations; hydroelectric reservoirs; oil, gas and coal extraction	Removes the trees that anchor soil and cycle nutrients; floods valley habitats; fragments the ecosystem with access roads and pipelines
Water	Boreholes and wells; river abstraction for irrigation; dams and water transfer schemes	Lowers the water table so deep-rooted trees die; reduces river flow downstream; concentrates livestock around new water points, stripping vegetation nearby
Materials and minerals	Timber; gold, bauxite and coal mining; quarrying	Clear-felling; toxic contamination of soils and rivers (for example mercury from gold extraction); permanent loss of soil structure

Ecosystem services: The benefits people obtain from ecosystems. These are usually grouped as *provisioning* (food, timber, water), *regulating* (carbon storage, flood control, pollination), *supporting* (soil formation, nutrient cycling) and *cultural* (recreation, spiritual and scientific value). Much ecosystem damage happens because provisioning services have a market price and the other three do not.

Exam Tip — Use Is Not the Same as Damage: Questions often ask you to explain how ecosystems are damaged by human activity. The strongest answers make the distinction above explicitly: identify a use, then identify the threshold at which that use stops being sustainable, and say what crosses it — usually population growth, market demand, or a technology that raises extraction rates.

2. Deforestation in the Tropical Rainforest

Deforestation: The permanent removal of trees from a forested area, converting land to non-forest use. The global rate is approximately **10 million hectares per year** (equivalent to losing a football pitch every 2–3 seconds).

Causes of Deforestation

Cause	Detail	Scale / Significance
Cattle ranching	Forest cleared for pasture — cheapest land use	Biggest single driver in the Amazon (most cleared land becomes cattle pasture)
Soya farming	Global demand for soya (livestock feed, biofuel)	Brazil is world's largest soya exporter; the 2006 Soy Moratorium cut direct clearance for soya in the Amazon, but soya expansion still pushes ranchers deeper into the forest
Commercial logging	Hardwoods (mahogany, teak) sold internationally	Legal and illegal logging opens up roads into intact forest

Mining	Iron ore, gold, bauxite extraction — Amazon has vast mineral wealth	Creates pollution, mercury contamination of rivers (illegal gold mining)
HEP dams	Large reservoirs flood vast forest areas	Belo Monte Dam (Brazil) flooded 500km ² of Amazon
Road building	Trans-Amazonian Highway opened up interior	Enables further settlement, farming, logging along new routes
Population growth	Land hunger — small farmers clear forest for subsistence	Migration to Amazon from overcrowded cities; landlessness
National debt	Developing countries exploit forests for export revenue to repay debt	Short-term economic pressure overrides environmental concerns

Consequences of Deforestation — in Detail

1. Biodiversity Loss

Habitat destruction is the single greatest driver of species extinction globally. Some scientists have estimated that as many as **137 plant, animal and insect species are lost every day** from rainforests worldwide — roughly one species every ten minutes. Because of the high interdependence of rainforest species, the extinction of one species can trigger a cascade of secondary extinctions. A 2015 study estimated that between a third and a half of Amazonian tree species could qualify as threatened with extinction because of deforestation. Many species are lost before they are even discovered or described by science.

2. Soil Erosion and Degradation

Without root systems binding the soil and leaf litter cushioning the impact of rain, intense tropical downpours dislodge and wash away topsoil at rates **up to 100 times faster** than in intact forest. The exposed laterite soil rapidly loses structure and fertility. Within 2–3 agricultural cycles, the land becomes essentially unusable for farming — yet it may take centuries for natural forest to regrow on degraded land. It has been estimated that around 2 billion hectares of deforested or degraded land worldwide could be restored.

3. Disruption of the Water Cycle

Trees recycle **50–80% of rainfall** through evapotranspiration. When a large area is deforested, regional rainfall can decline significantly. Studies suggest that heavily deforested parts of the Amazon already receive **noticeably less rainfall** and have longer dry seasons than forested areas. This creates a positive feedback: less forest = less rain = more drought stress on remaining trees = more forest die-off. Cities far from the Amazon, including São Paulo (population 22 million), receive significant proportions of their rainfall from Amazonian "flying rivers" — meaning deforestation in Brazil creates water shortages in cities hundreds of kilometres away.

4. Climate Change — Carbon Release

Tropical forests store approximately **250 billion tonnes of carbon** in their biomass and soils — equivalent to about 25 years of current global CO₂ emissions from fossil fuels. When forest is burned or cleared, this carbon is released as CO₂ and methane. Deforestation and land-use change currently account for approximately **10–15% of annual global greenhouse gas emissions**. Crucially, a degraded Amazon could transition from a carbon sink (absorbing CO₂) to a carbon source (emitting CO₂) — research published in 2021 in *Nature* showed that the south-eastern Amazon is already emitting more carbon than it absorbs, due to a combination of deforestation and climate-driven drought.

5. Displacement and Rights of Indigenous People

The Amazon is home to **hundreds of Indigenous peoples** (well over 1 million people), including dozens of uncontacted groups. Deforestation destroys not only their physical environment but their culture, traditional knowledge (including plant-based medicines), and survival systems. The Yanomami crisis of 2022–23, when 20,000 illegal gold miners (garimpeiros) invaded Yanomami territory, resulted in widespread malaria, mercury poisoning, malnutrition, and a dramatic rise in child mortality. The Brazilian health authority declared a public health emergency in January 2023.

6. Loss of Medicinal Potential

An estimated **25% of all modern pharmaceutical drugs** are derived from or inspired by rainforest plants. The muscle relaxant curare (used in anaesthesia) comes from Amazonian plants; quinine (malaria treatment) from Andean cinchona bark; and vincristine (childhood leukaemia treatment) from the rosy periwinkle of Madagascar. Given that less than 1% of known rainforest plant species have been scientifically tested for medicinal properties, the pharmaceutical potential that is being permanently destroyed is incalculable.

Exam Tip — Using Statistics Effectively: In 6–8 mark questions, examiners reward specific factual support. Always try to include at least one statistic per consequence you describe. Do not invent

statistics — use the ones you have memorised (up to 137 species/day; 250 billion tonnes carbon; 25% of medicines from rainforest plants). Quoting a study or named scientist (e.g. "research published in Nature, 2021") adds further credibility.

3. Case Studies — Rainforests Under Pressure

Case Study A: The Amazon Rainforest, Brazil

Location: South America, spanning 9 countries. Approximately 60% lies within Brazil. The Amazon River drains the basin — the world's largest river by discharge volume.

Size: ~5.5 million km² of tropical rainforest — roughly the size of Western Europe.

Biodiversity: Home to an estimated 10% of all Earth's species: 40,000 plant species, 1,300 bird species, 3,000 fish species, 430 mammal species.

Rate of loss: Approximately 17% of the forest has been cleared since the 1970s. Peak deforestation was in 2004 (~27,000 km²/year). After falling following policy intervention (Forest Code), it rose again after 2018 — under Bolsonaro's government (2019–2022), deforestation reached 10,000–13,000 km²/year (peaking at about 13,000 km² in 2021). Under Lula (from 2023) it fell sharply again: Brazil's national space agency INPE recorded 5,796 km² for the twelve months to July 2025 — an 11% fall on the previous year and the lowest annual figure since 2014.

Main causes in the Amazon: Cattle ranching (the largest driver), soya cultivation (often indirectly), logging, mining (particularly illegal gold mining in indigenous territories), road building.

Tipping Point: Scientists (particularly Carlos Nobre and Thomas Lovejoy) warn that if 20–25% of the Amazon is deforested, the forest may reach a 'tipping point' — reduced evapotranspiration causes regional drying that prevents forest regrowth. As of 2024, approximately 17–20% has been cleared and a further ~17% is degraded. Seasonal drought events driven by climate change have worsened — in 2023, an extreme drought reduced water levels on the Amazon River to historic lows, stranding river communities and killing more than 150 river dolphins in Lake Tefé; the 2024 drought brought even lower river levels.

Indigenous impacts: The Yanomami people (35,000 individuals) suffered a major humanitarian crisis in 2022–23 when 20,000 illegal gold miners (garimpeiros) invaded their territory, spreading malaria and mercury poisoning. Child deaths from malnutrition and disease rose sharply, and Brazil declared a public health emergency in January 2023.

Significance: The Amazon stores approximately 150–200 billion tonnes of carbon. Its loss would accelerate global climate change significantly and is considered by scientists to be one of the most critical planetary tipping points.

Case Study B: Borneo (SE Asia) — Deforestation for Palm Oil

Location: Borneo is the world's third largest island, divided between Indonesia (Kalimantan), Malaysia (Sabah and Sarawak), and Brunei. It lies roughly between 7°N and 4°S, straddling the equator, in the heart of the Indo-Malayan rainforest region.

Size of original forest: Borneo (about 740,000 km²) was once almost entirely covered by rainforest. It is home to some of the oldest rainforest on Earth — estimated at 130 million years old, significantly older than the Amazon.

Rate of loss: Between 1973 and 2010, Borneo lost approximately **30% of its forest cover** — about 170,000 km² (larger than England). Deforestation in the 2000s was one of the fastest rates on Earth at the time. The primary driver has been **palm oil plantations**, which now cover over 6 million hectares in Malaysian and Indonesian Borneo combined.

Main causes:

- **Palm oil:** Used in ~50% of all supermarket products globally (biscuits, shampoo, margarine, cosmetics). Palm oil is the highest-yielding vegetable oil and the cheapest to produce, making it extremely profitable to grow. Vast areas of peatland rainforest have been drained and burned to make way for plantations.
- **Commercial logging:** Tropical hardwoods (meranti, teak) exported primarily to Japan, China, and Europe. Both legal and illegal logging have been major drivers since the 1960s.
- **Mining:** Coal and bauxite mining in Kalimantan (Indonesian Borneo).
- **Transmigration:** The Indonesian government's policy of relocating people from densely populated Java to Borneo, placing pressure on land from subsistence farming.

Ecological consequences:

- **Orangutan:** The Bornean orangutan (*Pongo pygmaeus*) lost nearly **150,000 individuals between 1999 and 2015** — roughly half the population — primarily due to habitat loss and hunting. It is now classified as Critically Endangered (IUCN). Orangutans need large contiguous forest territories and cannot survive in plantation monocultures.

- **Bornean pygmy elephant and Sumatran rhino:** the elephant is Endangered and the rhino Critically Endangered (now extinct in the wild in Malaysian Borneo), with small fragmented populations increasingly unable to maintain viable breeding populations.
- **Peat fires:** Borneo's peatlands (carbon-rich waterlogged soils up to 20m deep) have been drained for plantations. When drained peat dries out, it becomes highly flammable. The 1997–98 El Niño-driven fires in Indonesia burned roughly 10 million hectares of land (much of it peat), releasing an estimated 0.8–2.6 billion tonnes of carbon — equivalent to 13–40% of global fossil fuel emissions for that year. Similar large-scale fires occurred in 2015 and 2019.
- **River pollution:** Palm oil mill effluent (POME) has severely polluted rivers in Sabah and Kalimantan, with dissolved oxygen levels in some rivers falling to near zero, wiping out freshwater fish populations.

Responses: The Roundtable on Sustainable Palm Oil (RSPO) was established in 2004 to certify sustainably produced palm oil. By 2023, approximately 19% of global palm oil was RSPO-certified. Critics argue the RSPO's standards are insufficiently strong and enforcement is weak. Some countries (Norway, UK) have committed to "deforestation-free" supply chains. Consumer pressure has led some companies (Unilever, Nestle) to commit to sourcing only certified palm oil — but compliance is difficult to verify across complex global supply chains.

Comparison with Amazon: Borneo's deforestation is driven more by a single global commodity (palm oil) than the Amazon's multiple drivers (cattle, soya, logging, mining). This potentially makes it easier to address through supply chain certification — but the global scale of palm oil use makes systemic change extremely challenging. Borneo's forests are also unique — they share very few species with the Amazon and are irreplaceable even if the Amazon were fully protected.

Exam Tip — Using Two Case Studies: If asked to "use a case study to support your answer," one well-chosen example with specific facts scores higher than a vague general statement. If the question says "with reference to examples," you can briefly compare both case studies — e.g. "In the Amazon, the main driver is cattle ranching, whereas in Borneo, palm oil for global consumer goods is the primary driver, affecting over 6 million hectares." Contrast shows geographical understanding of scale and variation.

4. Hot Semi-Arid Grassland Under Threat

The savanna is damaged more quietly than the rainforest. There is rarely a satellite image of a cleared block; instead the grass cover thins, bare soil appears between tussocks, woody scrub replaces palatable grasses, and the soil blows or washes away. Because the change is gradual it is easy to miss until it is difficult to reverse. Eduqas requires you to understand how human activity damages this biome and how the processes inside it are affected.

The Main Pressures

Human activity	What happens on the ground	Which ecosystem process is affected
Overgrazing	Herd sizes grow with human population, and veterinary care reduces livestock deaths, so more animals graze the same land. Grass is cropped faster than it regrows and the growing point at the base is damaged	Vegetation cover falls, so less litter reaches the soil and the nutrient cycle slows; roots no longer bind the soil
Loss of mobility	Fences, land subdivision, farm expansion and park boundaries stop pastoralists and wild herds from moving between seasonal pastures	Grazing pressure concentrates permanently in one place instead of rotating, removing the recovery period the grasses depend on
Conversion to cropland	Grassland is ploughed for cereals, cotton or biofuel crops, often on marginal land during a run of wetter years	Deep grass roots are replaced by shallow annual crops; soil is left bare between harvests; monoculture collapses the food web
Fuelwood and charcoal cutting	Trees and shrubs are cut for cooking fuel and for urban charcoal markets	Removes shade, removes deep roots that recycled nutrients from below the grass root zone, and removes the windbreak effect
Borehole drilling	New permanent water points allow livestock to stay year-round where they once could not	Creates a ring of completely bare, trampled ground — a "piosphere" — extending several kilometres from each borehole

Changed fire regimes	Fire is either suppressed entirely or set too frequently	Suppression lets woody scrub invade and shade out grass ("bush encroachment"); too-frequent fire kills tree seedlings and exhausts nutrients
Poaching and predator removal	Elephants, rhino and large predators are killed for ivory, horn or in retaliation for livestock losses	Elephants push over trees and keep grassland open; predators control herbivore numbers. Remove them and the vegetation structure changes

How the Damage Propagates

The sequence is worth learning as a chain, because exam questions ask for process, not lists:

1. Vegetation cover falls below the point where it shields the soil.
2. Rain now strikes bare ground directly. Raindrop impact seals the surface, infiltration falls and surface runoff rises.
3. Topsoil — the only nutrient-rich layer — is washed into gullies in the wet season and blown away in the dry.
4. With less organic matter, the soil holds less water, so the next drought bites harder and fewer plants germinate.
5. Less vegetation means less evapotranspiration, which can reduce local rainfall — a **positive feedback loop** that keeps the system degrading.

Where this runs to completion in a hot semi-arid region it is called **desertification**, covered in detail in 3.4.

Case Study — The Sahel, West Africa

Where: A belt roughly 5,000 km long across Senegal, Mauritania, Mali, Burkina Faso, Niger, Nigeria, Chad and Sudan, between the Sahara and the wetter savanna to the south.

Pressures: Rapid population growth, expansion of cropping onto marginal grazing land, larger livestock herds, and heavy demand for fuelwood and charcoal around growing towns. Rainfall is naturally unreliable and the region suffered severe droughts in the 1970s and 1980s.

Lake Chad: The lake contracted dramatically from around 25,000 km² in the 1960s to roughly 2,500 km² in the mid-1980s — a combination of drought and heavy abstraction for irrigation. Its extent has partly recovered since, reaching about 14,000 km² in 2013 and much larger seasonal maxima in

recent years, and researchers now argue the long-term decline has halted. The livelihoods that collapsed during the low decades, however, have not simply returned.

Consequences: Loss of grazing and cropland, falling crop yields, competition between pastoralists and farmers over the same shrinking resource, and displacement — environmental pressure feeding directly into conflict and migration.

Case Study — The Maasai Mara, Kenya

Where: The Kenyan end of the Serengeti–Mara ecosystem, made up of the government-run Maasai Mara National Reserve and a ring of community-owned land around it.

The pressure: From the early 2000s, communally held Maasai group ranches were privatised and subdivided into individual plots. Owners fenced their land to protect grass for their own cattle. Subdivision and fencing are now regarded as the largest structural threat to the ecosystem, because they cut the corridors that migrating herds and dispersing wildlife depend on, push pastoralists onto more marginal land, and increase human–wildlife conflict. Cropping has expanded onto former pastoral land, and the indigenous forest on the Mara's escarpments has been progressively cleared.

Why it matters ecologically: A 2021 census by the Wildlife Research and Training Institute and Kenya Wildlife Service found that over 83% of the Mara's wildlife was outside the national reserve, on community conservancies and dispersal land. Protecting only the reserve therefore protects a minority of the ecosystem — the land under most pressure is the land most of the wildlife actually uses.

5. The Effects of Damage — Local and Global

Eduqas asks specifically for the effects of ecosystem damage **at the local scale, including the effect on biodiversity, and at the global scale**. Keep the two separate in your answers.

Scale	Tropical rainforest	Hot semi-arid grassland
Local — soil	Nutrients held in biomass are removed with the trees; heavy rain leaches what is left; exposed soil can bake into laterite within a few years	Topsoil is lost to sheet wash and wind; surface crusting cuts infiltration; gullies develop at the start of the wet season

Local — water	Interception and evapotranspiration collapse, so runoff and flood peaks rise and dry season river flow falls	Water tables fall where boreholes over-abstract; springs and seasonal pools dry earlier each year
Local — biodiversity	Habitat is fragmented into islands too small to hold viable populations; specialists with narrow niches disappear first; edge effects dry out the remaining forest margin	Palatable grass species are grazed out and replaced by unpalatable or woody species; large mammals lose migration corridors; food webs simplify
Local — people	Indigenous communities lose land, food sources and cultural sites; mercury from gold mining contaminates rivers and fish	Falling yields and lost grazing drive food insecurity, competition between farmers and herders, and out-migration
Global — climate	Carbon stored in biomass is released; a major carbon sink is weakened; regional atmospheric moisture transport is disrupted	Carbon stored in soils and roots is oxidised; higher surface albedo over bare ground can suppress rainfall further
Global — biodiversity	Loss of a disproportionate share of world species — the rainforest holds over half of them on about 6% of the land surface	Loss of the world's last great herds of large mammals and of the open-country predators that depend on them
Global — people	Loss of potential medicines; disruption of global commodity supply chains; displacement and migration pressure	Reduced food production in already food-insecure regions; displacement and conflict with regional and international consequences

Exam Tip — Naming the Scale: If a question says "at different scales", signpost it: "At the local scale... At the global scale...". Examiners are looking for the shift, and a well-organised answer that names the scale explicitly moves up a band even where the content is similar to a weaker one.

6. Why Ecosystems Are Conserved and Managed

Before describing management strategies, be able to explain **why** ecosystems are protected at all. The reasons split into four groups, and a good evaluation question answer uses more than one.

- **Economic:** Ecosystems produce marketable goods — timber, fish, grazing, tourism income. A degraded ecosystem produces less of all of them. Tourism revenue is the single strongest economic argument for protecting East African savanna.
- **Environmental:** Carbon storage, water cycle regulation, flood control, pollination and soil formation are services that would be extraordinarily expensive to replace artificially.
- **Social and cultural:** Indigenous and local communities have lived in and shaped these ecosystems for generations; the land carries cultural and spiritual value as well as material value.
- **Ethical and scientific:** The argument that species have a right to exist independent of their usefulness, alongside the practical point that undiscovered species may hold future medicines.

Sustainable management: Managing an ecosystem so that it continues to meet people's needs today without reducing its ability to meet the needs of future generations. In practice this means extraction rates that do not exceed regeneration rates, and giving local people an economic reason to keep the ecosystem intact.

7. Sustainable Management of the Tropical Rainforest

Sustainable management aims to use the rainforest to meet present needs **without compromising the ability of future generations** to benefit from it.

Strategy	How it Works	Evaluation
Selective logging	Only mature trees of specific species are felled; the rest of the forest is left intact	Reduces biodiversity loss vs. clear-felling; but roads still open up forest; takes 50+ years for full recovery
Replanting / reforestation	New trees planted to replace those felled; often monocultures	Restores carbon storage; but monocultures have far less biodiversity than natural forest

Ecotourism	Tourists pay to visit and experience the forest; income supports conservation	Costa Rica: 26% of land protected; ecotourism worth \$3bn/year. However: can disturb wildlife; limited to accessible areas
Debt-for-nature swaps	A country's debt is reduced or cancelled in exchange for committing to protect a specified area of forest	Bolivia, Ecuador used this model. Effective when government has capacity to enforce; corruption can undermine it
National parks and reserves	Legal protection from development (e.g. Tumucumaque National Park, Brazil — world's largest tropical national park)	Effective if well-enforced; illegal logging/mining still occurs in many "protected" areas
CITES	International agreement banning trade in endangered species and protected timber (e.g. mahogany)	Reduces commercial incentive to log protected species; but enforcement difficult
Indigenous land rights	Legal recognition of indigenous peoples' rights to their traditional territories — they are proven best forest guardians	Indigenous territories in the Amazon have significantly lower deforestation rates; but politically contested
REDD+	UN scheme: developed countries pay developing countries to protect and restore forests as carbon offsets	\$billion-scale funding available; complex governance challenges; some criticism of "carbon offsetting" as greenwashing

Deeper Evaluation of Key Strategies

Indigenous Land Rights — the strongest evidence: Research comparing deforestation rates across different land tenure types in the Amazon consistently shows that indigenous territories have **dramatically lower deforestation rates** than equivalent non-protected forest or even formally designated national parks. Studies of satellite data have found that indigenous lands in Brazil have **much lower deforestation rates** than non-protected areas. Indigenous communities have the greatest incentive to protect their territories and have accumulated millennia of ecological knowledge about sustainable forest use. Yet these rights are frequently contested by governments and extractive industries — and under Bolsonaro, demarcation of new indigenous territories was halted entirely, directly contributing to increased deforestation.

REDD+ — promising but flawed: REDD+ (Reducing Emissions from Deforestation and Forest Degradation) is the most ambitious international mechanism, theoretically capable of channelling hundreds of billions of dollars from wealthy countries into forest protection. However, evaluation reveals significant weaknesses: (1) **additionality** — is the forest only protected because of REDD+ money, or would it have survived anyway? (2) **permanence** — carbon offsets from protected forest can be undermined if the forest is later burned; (3) **leakage** — protecting forest in one area may simply displace deforestation elsewhere; (4) a 2023 investigation found that over 90% of rainforest carbon credits issued by one major certifier (Verra) were "phantom credits" that did not represent real emissions reductions. This does not invalidate REDD+ entirely, but shows that robust monitoring and governance are essential.

Ecotourism — effective but limited scale: Costa Rica remains the most cited success story: forest cover recovered from 21% (1987) to over 52% (2023) and ecotourism contributes over \$3 billion/year to GDP. However, Costa Rica is a small, stable, middle-income country with good governance. Attempting to replicate this in the DRC (Congo Basin) — a larger, lower-income country with fragile governance and ongoing conflict — is far more challenging. Ecotourism also inherently requires visitor access, which limits its applicability to the most remote and ecologically valuable areas of the Amazon or Congo.

Consumer action and supply chains: Since most Amazon deforestation is driven by beef and soya production for global markets, consumer and corporate behaviour in importing countries can be powerful. The UK's Environment Act 2021 contains provisions requiring large companies to ensure their supply chains are free from illegal deforestation — one of the first such laws globally. The EU Deforestation Regulation (EUDR, agreed in 2023 but its start date has been postponed more than once) goes further, requiring companies to prove their products (beef, soya, palm oil, coffee, cocoa, rubber, timber) did not cause deforestation before they can be sold in the EU. These laws, if well-enforced, could significantly reduce the economic incentive for deforestation by cutting off market access for destructive producers.

Case Study: Ecotourism in Costa Rica

Costa Rica has transformed from one of the world's highest deforestation rates in the 1970s to a global leader in conservation. Today, 26% of its land area is protected. Ecotourism brings in over \$3 billion per year and employs 13% of the workforce. The government's **Payment for Ecosystem Services (PES)** scheme pays landowners to keep forest standing. Forest cover has recovered from 21% (1987) to over 52% (2023). This demonstrates that economic incentives aligned with conservation can be more effective than protection alone.

8. Sustainable Management of Hot Semi-Arid Grassland

Savanna management works differently from rainforest management. The rainforest problem is usually "how do we stop the clearing?"; the savanna problem is usually "how do we keep people and animals moving?" Almost every successful strategy restores mobility, rotation or recovery time in some form.

Strategy	How it works	Evaluation
Rotational grazing	Land is divided into blocks grazed in sequence, so each block gets months of rest before livestock return	Cheap, uses existing knowledge and restores the recovery period grasses need — but requires agreement between many landowners, which is exactly what subdivision destroys
Community conservancies	Neighbouring landowners pool their land into a single conservancy, agree to remove fences and limit livestock, and receive a monthly lease payment funded by tourism	Gives conservation a direct cash value to the people who bear its costs. Land under conservancy management around the Maasai Mara doubled to about 1,420 km ² in the decade after the Maasai Mara Wildlife Conservancies Association formed in 2013 — an area close to that of the national reserve itself. Weakness: income depends on tourist numbers, which collapse during crises, and critics argue lease terms can favour tour operators over landowners
Wildlife corridors	Strips of unfenced land negotiated between protected areas to reconnect migration routes	Directly addresses the fragmentation that causes most savanna wildlife loss; but needs long-term agreement from every landowner along the route
Controlled burning	Deliberate, patchy, early dry season fires set at low intensity	Mimics the natural fire regime, clears dead litter, releases nutrients and prevents the fuel build-up that causes destructive late-season fires; requires skill and can go wrong in a dry year
Agroforestry and farmer-managed natural regeneration	Trees are protected and encouraged to regrow within cropland rather than	Very low cost and locally driven; restores shade, nitrogen and deep nutrient cycling. Slower and less visible than planting schemes, so it attracts less funding than it deserves

	being cleared; roots are left in the ground to resprout	
Water point management	Boreholes are spaced deliberately, and some are closed seasonally, so livestock cannot occupy one area year-round	Prevents the ring of bare, trampled ground around permanent water; unpopular with herders in a drought
Anti-poaching and community rangers	Local people are employed and trained as rangers	Provides income and local legitimacy; protects the large mammals that shape the vegetation. Expensive to sustain and dependent on external funding

Case Study — The Great Green Wall, Sahel

What it is: An African Union initiative, supported by the UNCCD, to restore degraded land in a belt across the Sahel. The original image of a single wall of planted trees has been replaced by a mosaic approach — restoring land through agroforestry, natural regeneration, water harvesting and improved grazing rather than by planting a continuous strip.

Ambition: 100 million hectares of degraded land restored, 250 million tonnes of carbon sequestered and 10 million green jobs created by 2030.

Progress: Roughly 18 million hectares restored and around 350,000 jobs created so far, with more than US\$14 billion raised or pledged. The initiative's own assessment is that it is about 15% of the way to its 2030 target, and that reaching it would require restoring around 8.2 million hectares a year with roughly \$4.3 billion of annual funding.

Evaluation: The shift from "plant a wall" to "restore a mosaic" is widely seen as the right correction — early tree-planting attempts had very poor survival rates because seedlings were planted where rainfall could not support them. The remaining obstacles are not ecological but organisational: incomplete data, coordination between many governments and NGOs, insecurity in parts of the Sahel, and the lag between money being pledged and work happening on the ground.

Which Approaches Work Best?

- **Local, low-technology and low-cost strategies tend to outperform large engineered schemes** in this biome. Farmer-managed natural regeneration, stone lines that slow runoff and drip irrigation all work because they can be maintained by the communities using them without outside funding.

- **Strategies that pay local people succeed where strategies that only restrict them fail.** A conservancy lease payment competes with the income from fencing and cropping; a legal ban does not.
- **Scale is the persistent problem.** Individual projects demonstrably work; the difficulty is that degradation is occurring across millions of hectares and restoration is happening across thousands.

9. Exam-Level Evaluation

Exam Tip — Evaluating Management Strategies: Questions asking you to evaluate management strategies require you to go beyond description. Structure your answer using:

- **What the strategy does** (brief)
- **Why it is effective** (with evidence/example)
- **Why it might be limited or ineffective** (e.g. cost, enforcement, scale, political will)
- **Overall judgement** — which approaches are most/least effective and WHY

Generally, strategies that create economic value for local communities (ecotourism, indigenous rights, PES) tend to be more sustainable than top-down protection alone, because communities have an active incentive to protect the forest.

Key Argument for 8+ Mark Questions: The most effective management combines multiple approaches — local (indigenous rights + ecotourism), national (legal frameworks + enforcement), and international (REDD+, debt swaps, CITES, supply chain laws). No single strategy is sufficient. The underlying driver — the global demand for beef, soya, and timber — also needs to be addressed through consumer choices and trade agreements.

Exam Tip — Synoptic Links: Rainforests connect to several other topics across the Eduqas spec. Climate change (topic 2.4) is worsening droughts in the Amazon — link these explicitly. Development (topic 1.3) explains why countries like Brazil and Indonesia prioritise economic exploitation over conservation. Weather and climate (topic 2.3) — peat fires in Borneo are intensified by El Niño droughts. Pulling in these cross-topic links in 6–8 mark answers demonstrates the highest level of geographical thinking.

